Project Management for Engineering Teams

A practical guide for student project managers building a launchable rocket

Plan the work. Enable the team. Prove the result.

15–20 MINUTE TEAM BRIEFING

Introduction: Forbes Filip

Engineering, project leadership, and field deployment

- Bachelor of Science in Electrical Engineering
- Associate of Science in Electrical Engineering Technology
- 4+ years as an Electrical Engineering Technician
- 2+ years as an Engineering Project Lead
- 2+ years as an instructor of Electronics and Electrical Engineering Technology
- Two deployments to Antarctica as an Engineering Field Lead

What I've learned

Complex engineering succeeds when design, people, testing, logistics, and operations are managed as one system.

The goal today is to share practical tools—not add bureaucracy to your project.

Good PMs ask; they do not assume

Technical expertise matters—but no PM needs to know everything.

Consult

Bring the relevant engineers and leads into requirements, design, and staffing discussions.

Clarify

Ask the technical owner to explain the outcome, assumptions, interfaces, and acceptance criteria.

Confirm

Record who agreed, what was decided, who owns the work, and what remains unresolved.

Escalate

If the team cannot agree—or authority is unclear—bring the issue to the faculty advisors rather than deciding unilaterally.

A PM can manage a task effectively without being the person best qualified to perform it.

Project managers are not the bosses

Responsibility for coordination does not grant unilateral technical authority.

Your responsibility

- Facilitate discussions and decisions
- Keep tasks, owners, and deadlines visible
- Ask questions and resolve blockers
- Document agreements and follow up

Not your unilateral decision

- Redefine mission scope or requirements
- Choose a technical design or solution
- Assign specialists without consultation
- Override agreed engineering decisions

If the team cannot agree on requirements, scope, timelines, or task assignments, involve the faculty advisors. PMs coordinate the discussion; advisors help resolve disputes and clarify decisions.

Why project management matters

Coordinate

Bring airframe, aerodynamics, propulsion, recovery, avionics, simulations, and launch operations together.

Make trade-offs

Balance performance, safety, reliability, schedule, budget, and the team's available time.

Deliver evidence

Turn a promising design into a launchable rocket with credible analysis, testing, documentation, and readiness.

The project manager’s job is to:

Keep the team aligned on competition requirements and the mission

Make work, decisions, dependencies, and risks visible

Help the team resolve blockers early—without personally doing every task

Project management reduces confusion, late surprises, and avoidable rework.

Start with a shared definition of success

1

Competition rules

What is required, scored, restricted, or due?

2

Mission objective

What must the rocket and team accomplish?

3

Deliverables

What designs, hardware, analysis, and documents must exist?

4

Acceptance criteria

What evidence proves each requirement is met?

A requirement is useful when the team understands what it means, who owns it, and how it will be verified—not just when it appears in a document.

Translate the competition brief into clear work before committing to a design or schedule.

Break the rocket project into connected workstreams

Mission & requirements

Competition rules, mission success criteria

Airframe & aerodynamics

Structure, stability, CAD, aerodynamic analysis

Propulsion & recovery

Motor integration, recovery system, deployment

Avionics & electronics

Sensors, flight computer, telemetry, power

Simulation & testing

Models, ground tests, flight-readiness checks

Operations & documentation

Safety, launch procedures, reports, readiness

Every workstream needs an owner, clear outputs, dependencies, and a definition of done.

Four questions every project manager should ask

Ask frequently—not just when a deadline is missed.

1

Is your task properly defined?

Do we agree on the expected output, constraints, and what “done” means?

2

Do you need help?

Are you blocked, missing information, or waiting on another person or team?

3

Do you have enough time?

Is the estimate realistic given workload, dependencies, and the deadline?

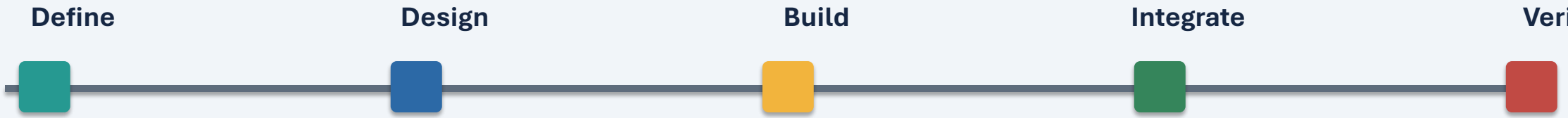
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Is there complete confidence when you say that you are done?

Has the result been checked, tested, and supported by evidence—not just attempted?

These questions reveal unclear work, blockers, unrealistic schedules, and false confidence while there is still time to act.

Manage dependencies, schedule, and risk



Dependencies

Late decisions or parts can block several teams. Identify the chain early.

Integration

Agree on interfaces and test across workstreams early—not only at the end.

Risk response

Name the risk, impact, mitigation, owner, and decision date.

END-TO-END EXAMPLE: RECOVERY SYSTEM

1 Define
Recovery must deploy reliably.

→

2 Plan
Assign an owner, test date, and criteria.

→

3 Monitor
Check progress and confirm parts are ready.

→

4 Respond
If delayed, identify blocker and backup plan.

→

5 Verify
Run the test, record results, fix and retest.

The loop repeats: update the plan when reality changes, and use evidence—not assumptions—to close the task.

Track outcomes—not just activity

Activity is not completion

“Worked on stability analysis” describes effort, not a reviewed result.

Define the evidence

Analysis reviewed, assumptions recorded, and findings connected to design decisions.

Use simple status

Not started • In progress • Blocked • Done—with evidence.

A useful project dashboard shows:

- Milestones on track, at risk, or late
- Open blockers and decisions needed
- Requirements verified versus outstanding
- Critical tasks due next
- Risks, owners, and response plans
- Test failures, fixes, and retests

A launchable rocket requires a verified system and a team ready to explain its decisions and evidence.

A lightweight operating rhythm

Weekly	Review milestones, workload, risks, and decisions.
At task start	Confirm output, owner, estimate, dependencies, and definition of done.
When blocked	Name the blocker, identify help or a decision needed, and agree on next steps.
At completion	Check acceptance criteria, record evidence, and communicate status.

Keep the process lightweight: add structure where it prevents confusion, risk, or rework.

The project manager’s day-to-day job

Monitor progress

Check the board and upcoming milestones. Ask what changed, what is next, what is blocked, and whether dates still look realistic.

Know team availability

Keep an active picture of each person's workload, class/exam periods, work hours, planned absences, and other commitments. Re-plan early when capacity changes.

Capture the meeting

Record attendance, key decisions and rationale, action items, owners, due dates, blockers, and unresolved questions. Share notes where the team can find them.

Make people successful

Check that each person has the information, tools, access, materials, decisions, and support needed to do their work. Use the four questions to uncover gaps.

Simple rule: if it affects the plan, a person’s ability to work, or a decision the team needs, make it visible and follow up.

Discussion: What will keep our rocket project on target?

1. Which competition requirements are non-negotiable?
2. Which interfaces or dependencies could surprise us?
3. What must be demonstrated—not merely completed?
4. Which of the four questions should we ask at our next check-in?

Your job is not to predict everything. It is to help the team respond well.